



• Introduction

The **GL-StarterKit v2** development board is designed as an educational hardware platform for embedded systems programming. Its primary goal is to help students and developers master real-time microcontroller development, bridging the gap between high-level software abstraction and low-level hardware control.

The board provides flexibility in teaching methodologies, supporting both hands-on development with bare-metal register manipulation and rapid prototyping using standard abstraction frameworks.

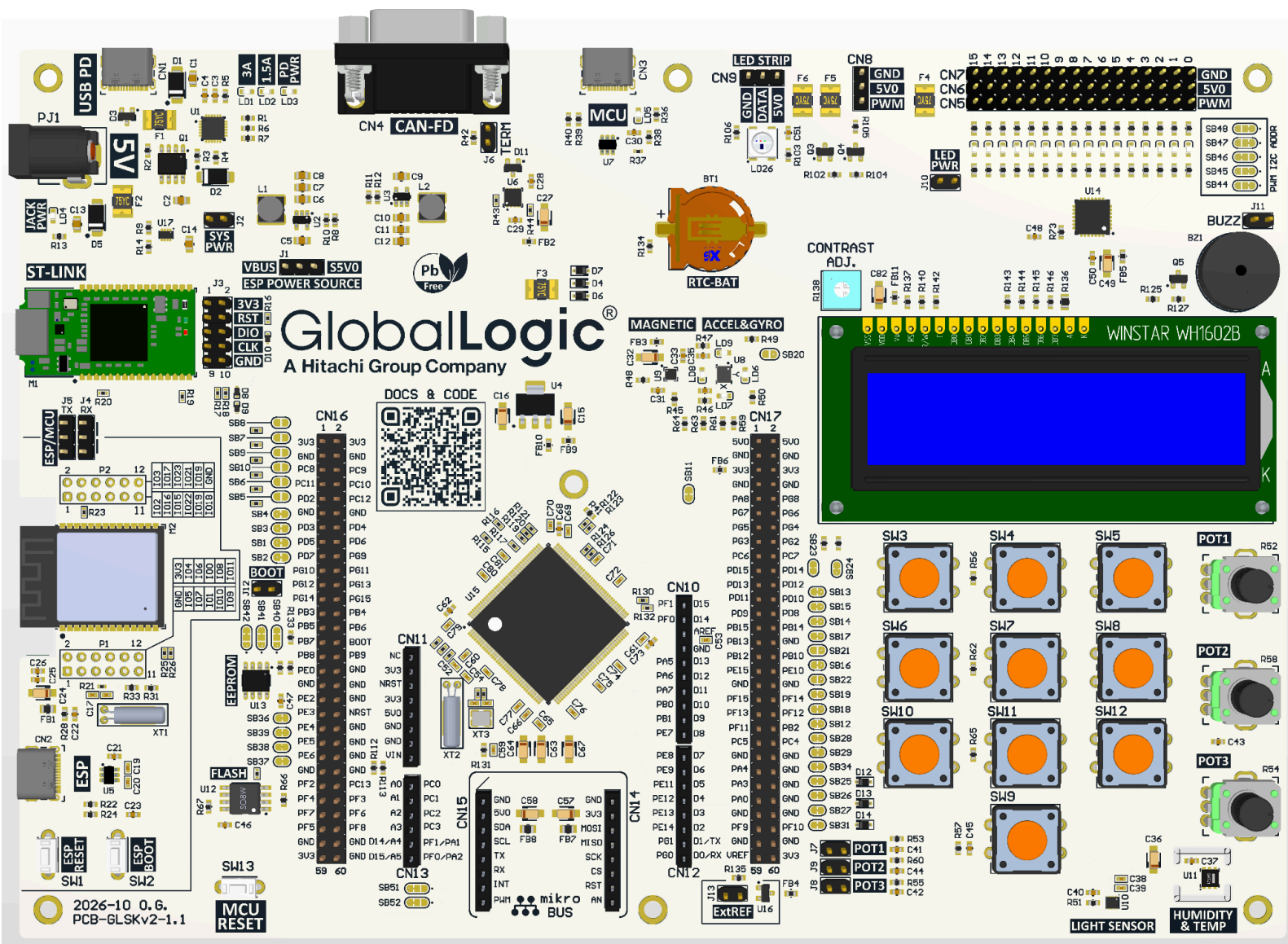
• Key Educational & Technical Capabilities

By working with the GL-StarterKit v2 platform, students will gain practical experience in:

- **Core Digital I/O & Peripherals:** Configuring GPIOs, Hardware Timers, Interrupts (EXTI), and Direct Memory Access (DMA).
 - **Communication Protocols:** Implementing low-level drivers and data transfer routines for **UART**, **I2C**, **SPI**, **SDIO**, and **USB**.
 - **Analog Data Acquisition:** Utilizing the internal high-speed ADC for precise analog measurements, sensor sampling, and signal processing.
 - **External Hardware Integration:** Interfacing with external memory modules, sensors, physical push-buttons, and character/graphical displays.
 - **System Modular Expansion:** Working with custom expansion shields and add-on boards compatible with the GL-SKv2 ecosystem to extend hardware functionality.
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Board Overview

- **Board Overview**



Key features

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MCU: [STM32H562ZG](#)

- High-performance 250 MHz CPU, ARM Cortex-M33 with TrustZone, 1-MByte Flash, 640-Kbytes RAM

Debugging and Flashing:

- Built in StLink v3 module ([STLINK-V3MODS](#))
- Debug serial UART port (J4, J5)
- Flashing with [STM32CubeProgrammer](#) via USB, using built in MCU bootloader (CN3, J12 - BOOT)

Power:

- Power Jack J1 (2.5x5.5mm), 5V 2.5A power supply input
- USB-C PD connector CN1, supports Power Delivery protocols (5V up to 3A), based on [STUSB4500L](#)

On board memory:

- 2Mb (16 MBit) Serial NOR FLASH (W25Q16J)
- 2Kb (16 KBit) Serial EEPROM (AT24C16C)

Sensors:

- 3-axis accelerometer ([LSM6DSO](#))
- 3-axis gyroscope ([LSM6DSO](#))
- 3-axis magnetometer/compass ([LIS2MDL](#))
- Relative humidity and temperature sensors ([HTU21D](#))
- Digital RGB, IR and Ambient Light Sensor ([APDS-9250](#))

Connectivity:

- Wi-Fi 6 in 2.4 GHz band, Bluetooth 5, Zigbee 3.0 and Thread 1.3 ([ESP32-C6-WROOM-1-N8](#) module)
- CAN FD port (industrial DB9 male connector) CN4

Display:

- On board 16x2 Character LCD Display Module ([WH1602B](#))
- Supporting [Waveshare 2.8inch TFT Touch Shield](#) (optional, external module)

17 PWM channels:

- 16x channels via PWM/LED controller [PCA9685](#) (CN5, CN6, CN7)
- 1x direct PWM channel from MCU (CN8)

Key features

IO Extension:

- 2x 60 pins IO expansion headers (CN16, CN17)
- [mikroBUS](#)™ socket (CN14, CN15)
- Arduino UNO compatible connector (CN10, CN11, CN12, CN13)

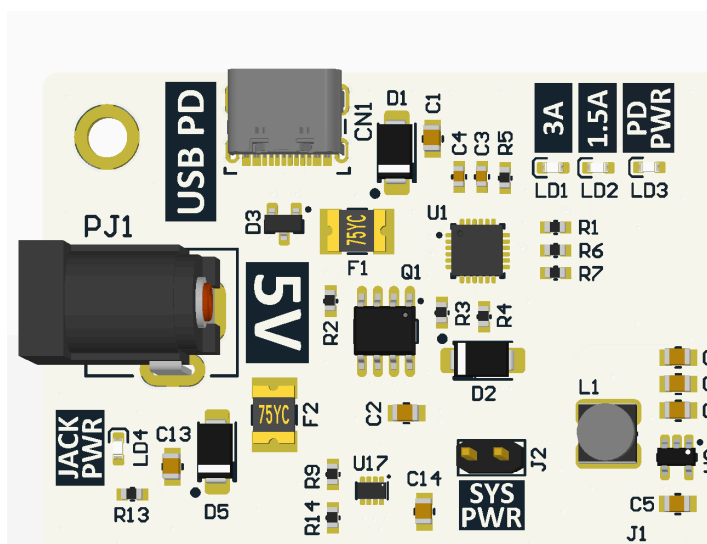
Other:

- 1x connector for LED strip with digital addressing (WS281x) (CN9)
- 1x Buzzer
- 3x analog on board Potentiometers
- 3x3 matrix keyboard (SW3-SW8, SW10-SW12)
- 1x general purpose button with wakeup compatibility (SW9)
- RTC backup battery 1220 socket

- **Getting Started**

Power Supply Setup

The **GL-StarterKit v2** board features a flexible and protected dual-input power supply system. It can be powered via an external DC adapter or a USB Type-C Power Delivery (PD) source.



Power supply connectors J1, CN1, and diagnostic LEDs.

Power Supply Inputs

The board provides two dedicated power connectors:

1. **DC Power Jack (PJ1)**
Nominal Input Voltage: 5V DC
Maximum Current: Up to 2.5A
Protection: Integrated reverse-polarity protection and overvoltage protection ($> 5V$).
Power Indicator: “LD4” (Green) illuminates when valid 5V power is present from the DC adapter connected to “PJ1”.
2. **USB Type-C PD Connector (CN1)**
Power Profile: 5V fixed profile, supporting current loads up to 3.0A.
Protection: Standard USB Type-C overcurrent and ESD protection. Additionally overvoltage protection ($> 5V$) at the output.

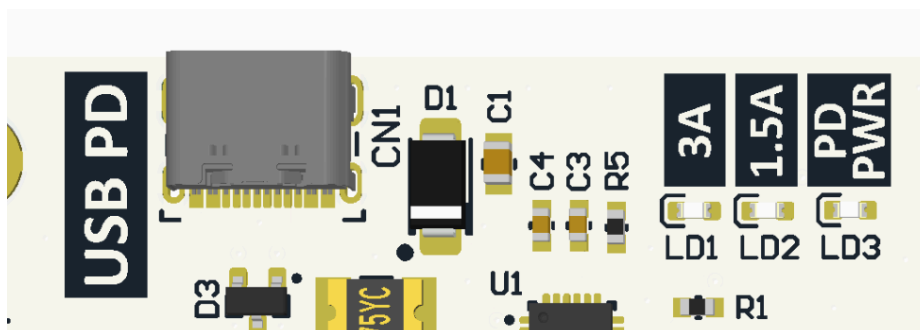
Power Priority Management

The board includes automatic power-path selection hardware:

- If both power sources “**PJ1**” and “**CN1**” are plugged in simultaneously, the **DC Power Jack “PJ1**” takes priority.
- The power multiplexer automatically switches the internal power rail to “**PJ1**” without resetting the MCU or interrupting board operation.

Power Diagnostic LEDs

The USB Type-C PD input “**CN1**” is equipped with three diagnostic LEDs **LD1**, **LD2**, **LD3**, to indicate the power capability offered by the connected USB-C host/adaptor:



LD1 (Red)

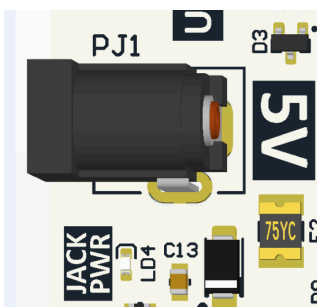
Illuminates when the connected USB PD supply supports high-current mode (5V/3.0A).

LD2 (Yellow)

Illuminates when the connected USB PD supply supports standard-current mode (5V/1.5A).

LD3 (Green)

Illuminates when valid 5V bus power is detected on the USB Type-C connector “**CN1**”.



LD4 (Green)

Illuminates when valid 5V power is present on the DC Jack “**PJ1**”.

First Power-Up Checklist

1. Connect your chosen power source (5V DC Adapter to “PJ1” or USB-C PD to “CN1”)
2. Verify that **LD3** (USB PD power) or **LD4** (DC Jack power) turns **Green**.
3. Check **LD1** or **LD2** to verify the available USB PD current limit for your application/shields.
4. Check for jumper **J2**, it must be **SHORTED** (installed)



5. If everything is powered correctly the 5V (S5V0, M5V0) and 3.3V (S3V3, M3V3) onboard rails will stabilize and power up the STM32H562ZG MCU and rest of the on board periferia.

Factory Test & Default Firmware Behavior

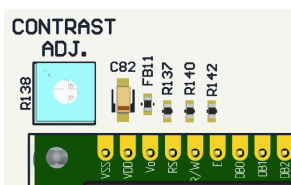
Upon the first power-up, the MCU automatically runs the factory test firmware pre-flashed during board manufacturing and verification.

Initial Startup Behavior:

1. **Display Contrast Adjustment** (First Power-Up):

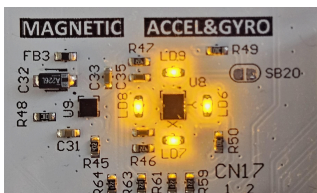
If the character display appears blank, too dim, or completely black (solid blocks) upon initial boot, you may need to adjust the LCD contrast balance:

Contrast Trimmer: Locate potentiometer **R138** positioned near the character display header.



Adjustment Procedure: Using a small screwdriver, gently rotate **R138** until the ready message and test text become crisp and clearly readable.

2. **Accel&Gyro LEDs:** Four Yellow LEDs **LD6, LD7, LD8, LD9** will illuminate immediately upon successful boot.



3. **PWM color LEDs:** During the **first power on**, the row of color LEDs located near the PWM headers **CN5, CN6, CN7** will light up if **jumper J10** is installed to power the LEDs.



4. **Character Display:** The onboard character LCD will initialize and display a readiness message indicating that the factory self-test suite is active and ready for input/testing.



Factory test firmware startup screen.

Firmware Flashing via MCU USB Boot Mode (DFU)

The STM32H562 microcontroller features a built-in ROM bootloader that enables firmware flashing directly via USB without needing an external hardware programmer (such as ST-LINK or J-Link).

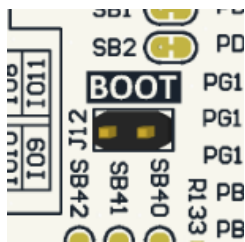
Prerequisite: Ensure that **STM32CubeProgrammer** (available from STMicroelectronics) is installed on your host system before proceeding.

Step-by-Step Flashing Procedure:

1. Set Boot Mode Jumper

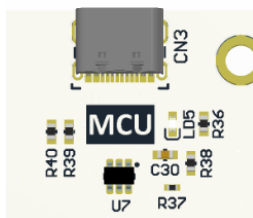
Close (short) the **BOOT** jumper **J12** on the board.

This forces the STM32H562 MCU to enter its internal ROM bootloader mode upon reset instead of booting from user Flash.



2. Connect USB Cable

Plug a USB cable into the MCU USB connector (**CN3**) and connect it directly to your host computer.



3. Reset MCU

Press and release the MCU RESET button (**SW13**) to reboot the microcontroller into USB DFU mode.



1. Open STM32CubeProgrammer

Launch the **STM32CubeProgrammer** application on your host PC.

2. Load Firmware File

Click on **Open file** and select your compiled binary or hex file (**.bin** or **.hex**).

3. Select USB Port

In the right-hand panel, set the interface drop-down menu to **USB**.

4. Connect to MCU

Click the Connect button. **STM32CubeProgrammer** will detect the board and display the device target memory address.

5. Flash Firmware

Click the **Download** button to begin flashing. Wait for the success confirmation dialog ("File download complete").

6. Disconnect Hardware

Disconnect from the board in **STM32CubeProgrammer** and close the application.
Remove the BOOT jumper **J12** and unplug the USB cable from connector **CN3**.

7. Boot New Firmware

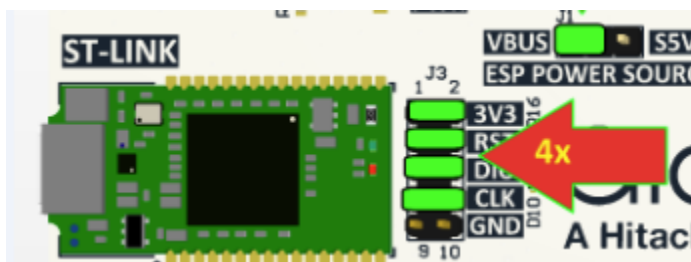
Press the MCU RESET button (**SW13**) to boot into your newly flashed application code.

Firmware Flashing & Debugging via Onboard ST-LINK/V3 Module

The **GL-StarterKit v2** board features an integrated **ST-LINK/V3** debugger module **M1**. This allows full hardware debugging, breakpoint control, and high-speed firmware flashing directly from your PC without needing any external debug probe.

ST-LINK/V3 Configuration **J3** Jumpers are located on the board - connection to MCU.

To route the debug signals (SWDIO, SWCLK, NRST) from the onboard ST-LINK/V3 module **M1** to the STM32H562 MCU, make sure the 4-pin jumper block **J3** is populated.



Step-by-Step Debugging & Flashing Procedure

1. Enable Onboard Debugger

Ensure all 4 jumpers on block **J3** are **closed** (installed). This connects the onboard ST-LINK/V3 (**M1**) signals directly to the target MCU.

2. Connect ST-LINK to PC

Plug a USB cable into the ST-LINK/V3 module USB connector (**M1**) and connect it to your host computer.

3. Verify Board Power

Ensure the board is powered up. The ST-LINK status LED on module **M1** will illuminate to indicate an active USB connection.

4. Launch Debug / Download

In your IDE (e.g., STM32CubeIDE, Keil MDK, VS Code, or OpenOCD), select ST-LINK (SWD) as the debug interface. Click Debug or Run. The toolchain will automatically detect the onboard ST-LINK/V3, flash the binary, and start real-time debugging.

- **Peripheral & Driver Examples (Hardware Test Suite)**

To assist with initial hardware verification and low-level driver development, a complete **Hardware Test Firmware** “examples/hw_test” is provided.

This project serves two purposes:

1. **Factory Diagnostics:** Used during board manufacturing to execute BIST (Built-In Self-Test) sequences.
2. **Educational Codebase:** Provides clean, bare-register driver implementations for onboard sensors, displays, and internal STM32H562 peripherals.

Project Structure Overview

The project is structured into three main layers:

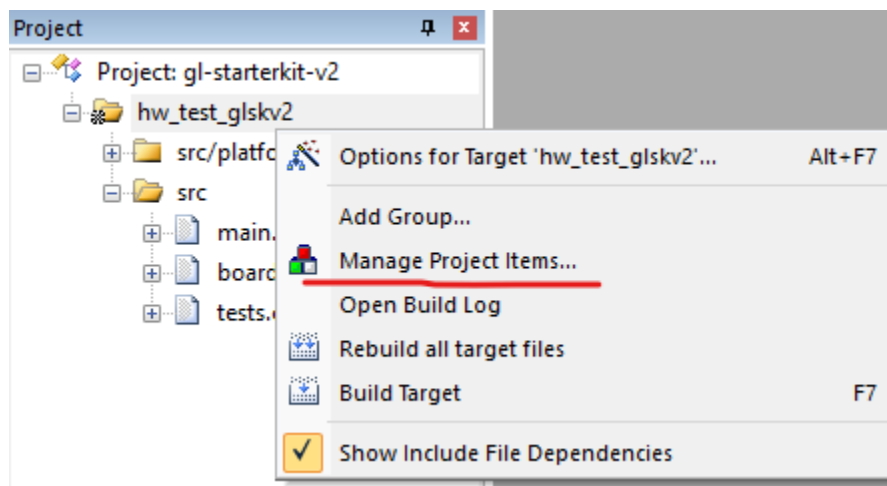
Application logic, Generic Driver interfaces, and MCU-specific Bare-Register implementations.

examples/0_mf_test/

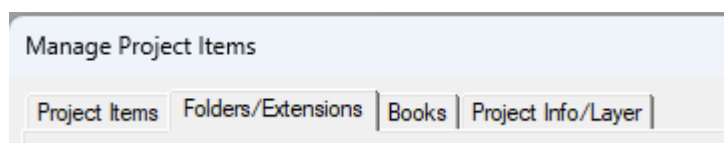
— gl-starterkit-v2.uvprojx	# Keil MDK Project File
— src/	
— main.c	# Main entry point
— board.c	# Board-level declarations and implementations
— tests.c / tests.h	# BIST loop and test suite implementations
— platform/	# Hardware drivers and MCU abstraction
— stm32h562xx/	# Bare-register peripheral drivers for STM32H562
— clock.c	# RCC / System Clock configuration
— gpio.c	# Low-level GPIO register setup
— i2c.c / spi.c	# Hardware bus controllers
— adc.c	# Analog-to-Digital Converter routines
— pwm_tim.c	# Timer-based PWM drivers
— irq.c	# IRQ Vector table
— linker.ld	# Linker script
— startup.c	# Startup code
— hd44780.c / .h	# Character Display (HD44780 controller)
— lis2mdl.c / .h	# LIS2MDL 3-axis Magnetometer (SPI)
— lsm6dso.c / .h	# LSM6DSO IMU Sensor (SPI)
— htu21d.c / .h	# HTU21D Humidity & Temp Sensor (I2C)
— apds9250.c / .h	# APDS-9250 Light Sensor (I2C)
— eeprom.c / .h	# External I2C EEPROM Driver
— flash_nor.c / .h	# NOR Flash storage memory driver
— gpio_keypad.c / .h	# Matrix Keypad scanning driver
— led_ws2812.c / .h	# Addressable RGB LED strip driver
— pwm_pca9685.c / .h	# PCA9685 PWM Controller driver

Building and Running the Test Firmware

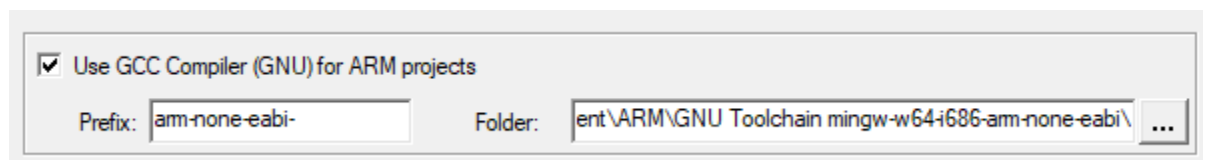
1. Download and install build toolchain from:
<https://gitlab.arm.com/tooling/gnu-toolchains-for-arm/-/tree/releases/15.2.rel1>
2. Open `gl-starterkit-v2.uvprojx` in **Keil MDK-ARM**.
3. Go to “**Manage Project Items**” via the dropdown menu on target name “**hw_test_glskv2**”.



Select tab “Folders/Extensions”.



On this tab should be set the checkbox “**Use GCC compiler (GNU) for ARM projects**” and set path to installed toolchain.



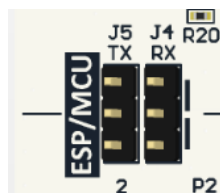
4. Select target `gl-starterkit-v2`.
5. Click **Rebuild (F7)** to compile all bare-register platform files and test suites.
6. Ensure the onboard ST-LINK/V3 jumpers (**J3**) are installed.
7. Click **Download (F8)** to flash the hardware test binary to the STM32H562 target.

- **Onboard Peripherals & Hardware Configuration**

The **GL-StarterKit v2** board integrates a rich set of user peripherals, communication interfaces, and configurable hardware options. This section details the operation and configuration of these onboard systems.

Debug UART Interface (VCP)

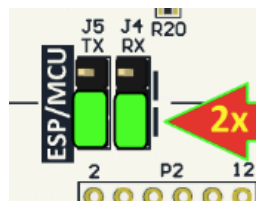
The Virtual COM Port (VCP) feature provided by the onboard **ST-LINK/V3** module allows direct serial logging and debugging via USB.



Route Selection (J4 & J5):

The RX and TX lines of the ST-LINK VCP can be dynamically routed to either the primary target MCU STM32H562 or the secondary wireless co-processor **ESP32-C6** using jumper headers **J4** and **J5**.

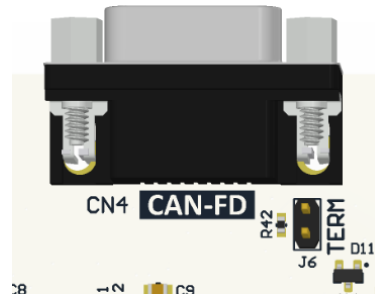
J4 routes RX line, **J5** routes TX line



Default configuration routes RxTx lines from/to **ST-LINK VCP** <> **ESP32-C6**

Industrial CAN FD Interface

Connector **CN4** provides access to the high-speed Controller Area Network with Flexible Data-Rate **CAN FD** controller.



Pinout Standard:

Follows the industrial CAN standard layout (CAN_H, CAN_L, GND).

Application:

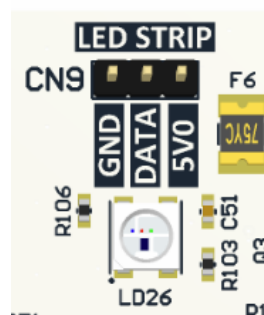
Ideal for teaching industrial automation buses, automotive networking, and differential bus signal dynamics.

If necessary, the **J6** terminating jumper must be installed.

RGB LED Strip Expansion

Connector **CN9** drives external addressable RGB LED strips (e.g., WS2812B / Neopixel) using a high-speed Timer + DMA signal generation routine.

Onboard First Pixel: The first addressable RGB LED (**LD26**) is mounted directly on the PCB. Additional external LED strips attached to **CN9** daisy-chain continuously from **LD26**.

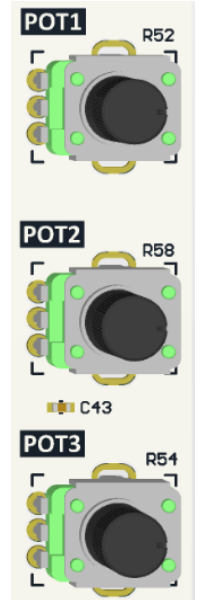
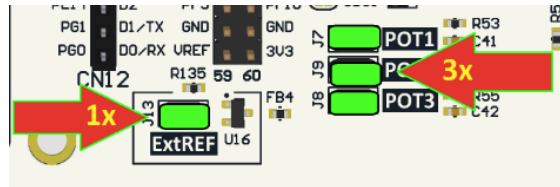


Precision Analog Measurement & Potentiometers

The board features three user potentiometers **POT1**, **POT2**, and **POT3** for studying analog-to-digital conversion (*ADC*).

For correct ADC operation, verify the following jumpers are populated (closed):

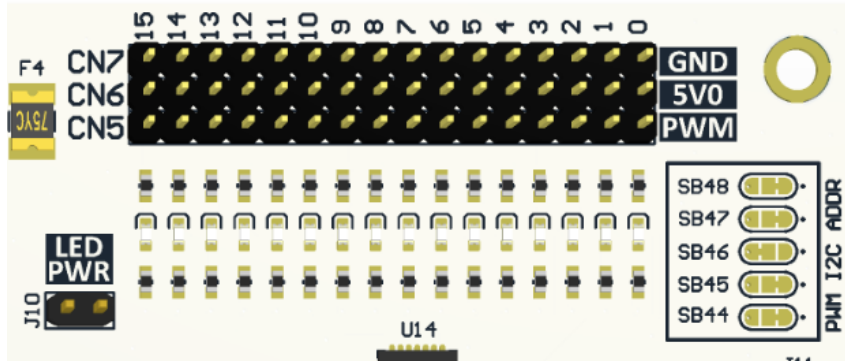
1. **J13 (Voltage Reference Jumper)**: Connects the ultra-stable 2.495V analog reference generated by the onboard TL431 IC (U16) to the MCU V_{REF+} domain.
2. **J7, J8, J9 (Analog Input Jumpers)**: Connects **POT1**, **POT2**, and **POT3** wiper pins directly to the MCU ADC channels.



Important: All four jumpers **J7**, **J8**, **J9** and **J13** must be installed for potentiometer sampling and precision ADC measurements to function.

16-Channel PWM LED Controller (PCA9685)

The board integrates a 16-channel I^2C -controlled PWM expansion driver connected to an array of individual status LEDs (LD10 through LD25).



LED Power Enable Jumper: The power rail for **LD10–LD25** is routed through jumper **J10**. To reduce power consumption or disable visual light output during operation, open (remove) jumper **J10**.

Hardware I^2C Addressing: The I^2C slave address of the PCA9685 controller can be reconfigured by soldering/unsoldering solder bridges **SB44**, **SB45**, **SB46**, **SB47**, and **SB48**.

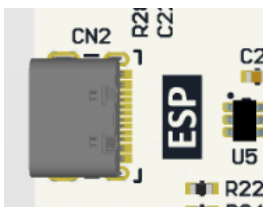
Direct MCU PWM Output

Connector **CN8** exposes a dedicated hardware PWM line driven directly by an internal STM32H562 timer.



Integrated Wireless Module (ESP32-C6-WROOM-1-N8)

The onboard **ESP32-C6** module provides Wi-Fi 6 and Bluetooth 5 (LE) capabilities to the platform. Programming and debugging the ESP32-C6 is handled via its dedicated USB connector **CN2**.



Power Source Selection

The ESP32-C6 module power input must be configured using jumper **J1**:



VBUS Position: Use this setting when the board is powered exclusively via the dedicated ESP32 USB connector **CN2** without an external power supply.

S5V0 Position: Use this setting when the main board is powered via the DC Power Jack **PJ1** or main USB-C PD input **CN1**.

Entering ESP32 Firmware Flashing Mode

1. Configure ESP Power

Set jumper **J1** to **VBUS** (if powering via **CN2**) or **S5V0** (if main power is active).

2. Connect USB Cable

Plug a USB cable into the ESP USB port (**CN2**) and attach it to your host computer.

3. Execute Boot Reset Sequence

Press and hold the **ESP BOOT** button (**SW2**). While holding **SW2**, press and release the **ESP RESET** button (**SW1**), then release **SW2**.

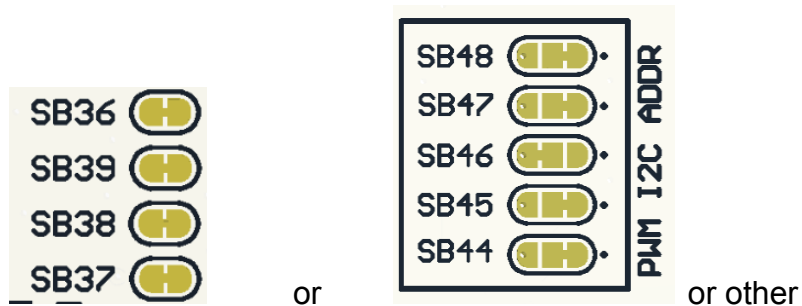
4. Verify Flash Readiness

Open the ST-LINK VCP UART console in your terminal (115200 baud). The ESP32-C6 will print a startup message indicating it is waiting for download mode. Proceed with flashing via **esptool** or ESP-IDF.

Factory Solder Bridge Configuration (SBx)

Caution: Several solder bridges (SBx) are shorted (closed) by default from the factory using thin PCB copper traces.

For example:



To reconfigure these buses or isolate specific peripherals, carefully cut the thin trace between the pads using a PCB knife before soldering new bridge configurations.